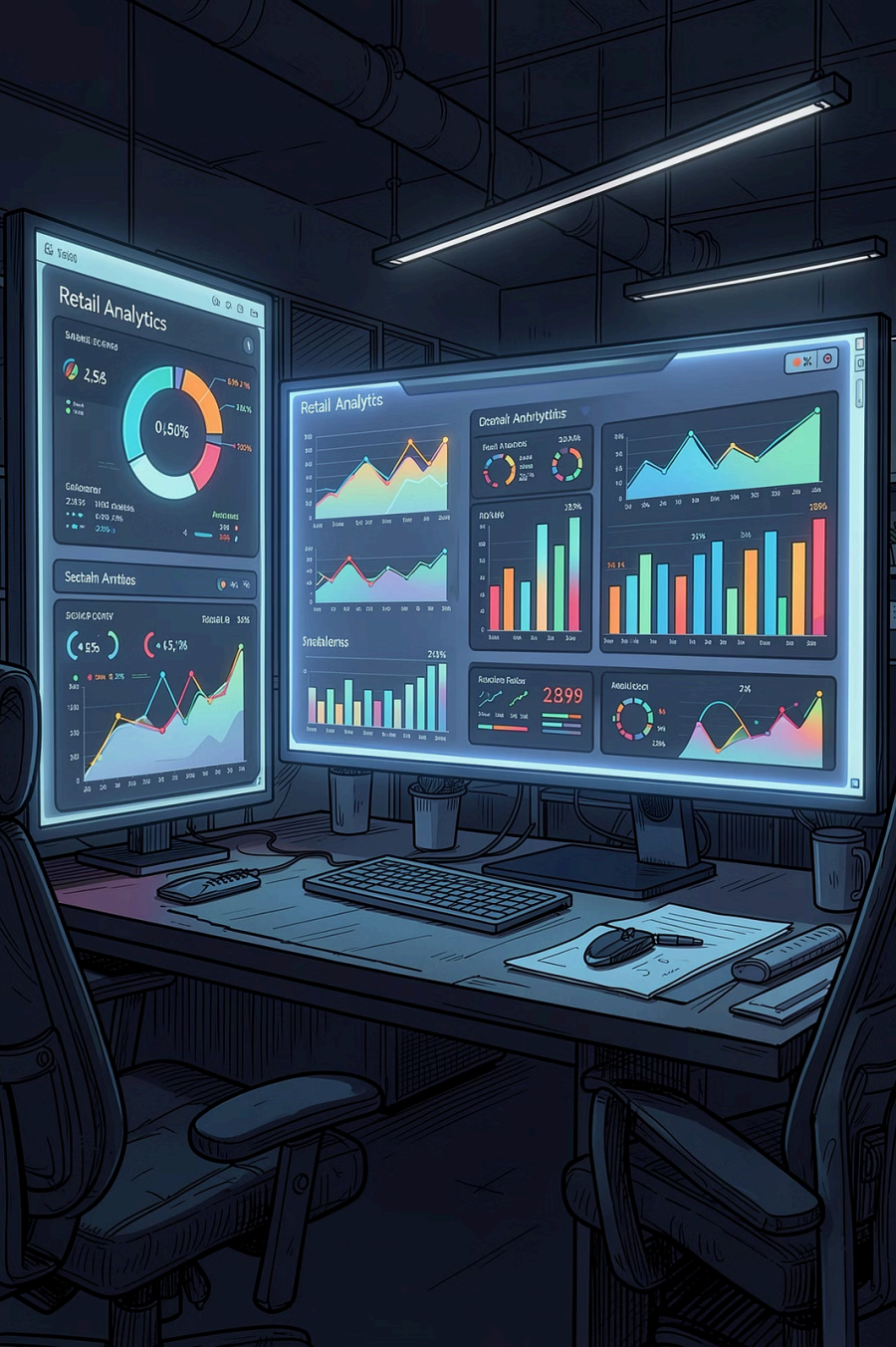




Customer Shopping Behavior Analysis

3,900 transactions. 18 features. One goal — uncover what drives purchase decisions and revenue.

RETAIL ANALYTICS

PYTHON · SQL · POWER BI

Dataset at a Glance

3,900

Transactions

Across all product categories

18

Features

Demographics, behavior, purchase details

\$59.76

Avg. Purchase

Mean spend per transaction

37

Missing Values

In Review Rating — imputed by category median

Data Preparation Pipeline

1

Load & Explore

pandas: `df.info()`, `.describe()`

2

Clean

Impute missing ratings, snake_case columns

3

Engineer

age_group bins, purchase_frequency_days

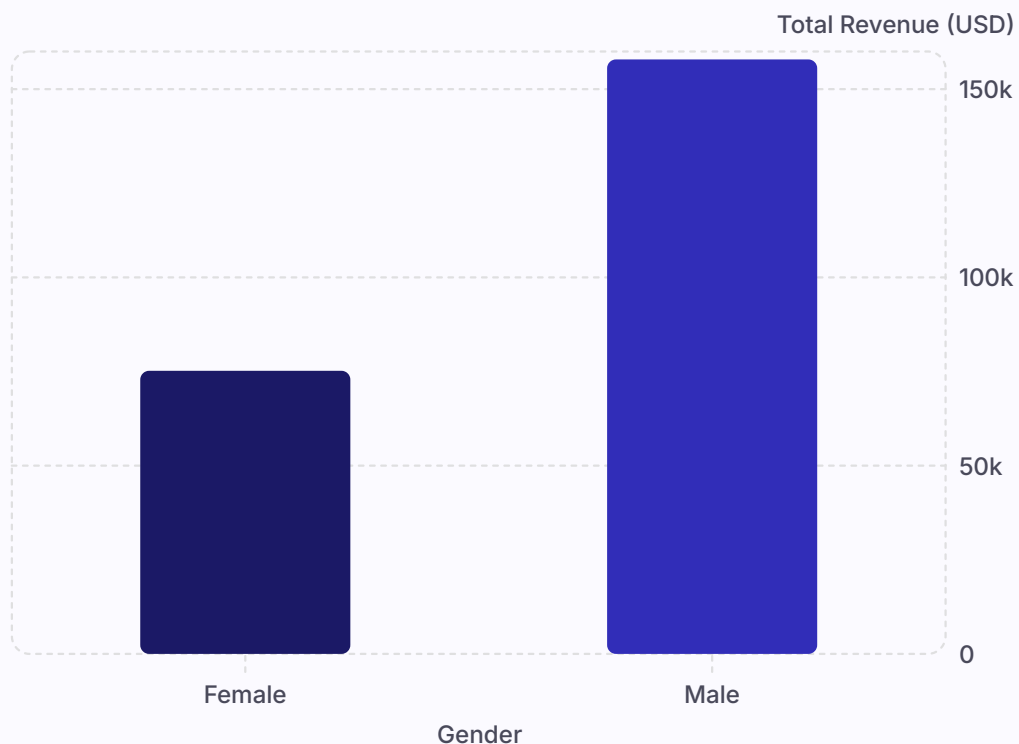
4

Load

Pushed to PostgreSQL for SQL analysis



Revenue: Male vs. Female



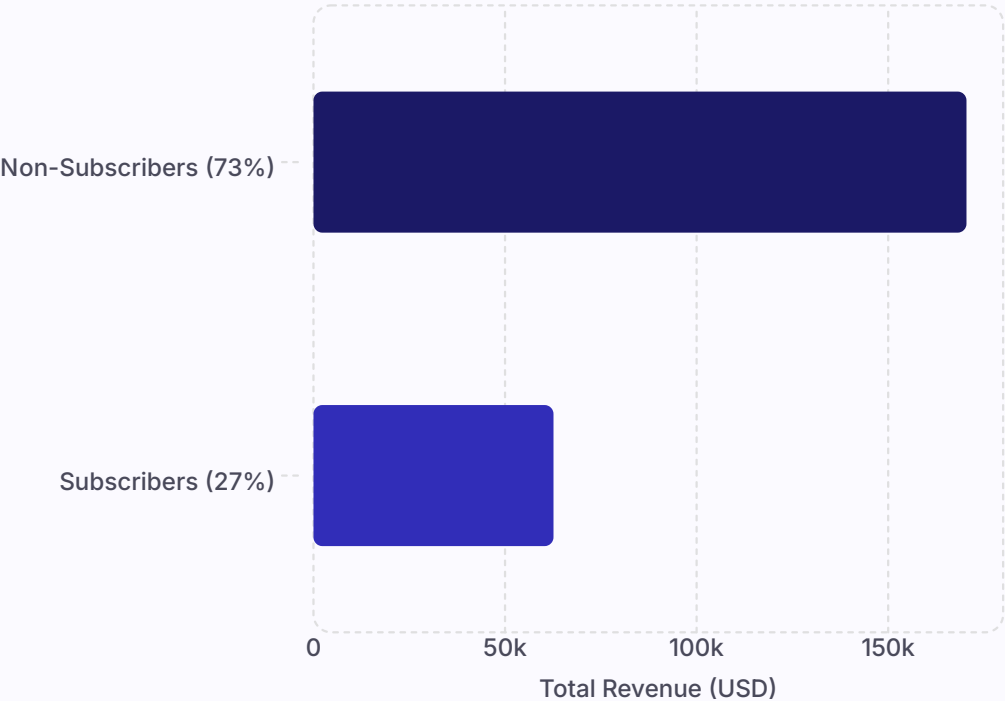
Male customers drive 2× more revenue

Despite females representing a significant portion of buyers, male customers generated **\$157,890** vs. **\$75,191** — a critical gap for targeting strategy.

📌 2,652 male customers vs. 1,248 female customers — both volume and spend favor the male segment.

Subscribers vs. Non-Subscribers

Subscription Status



Subscribers are underpenetrated

Only **27%** of customers subscribe, yet average spend is nearly identical — **\$59.49** vs. **\$59.87**. The opportunity is conversion, not upsell.

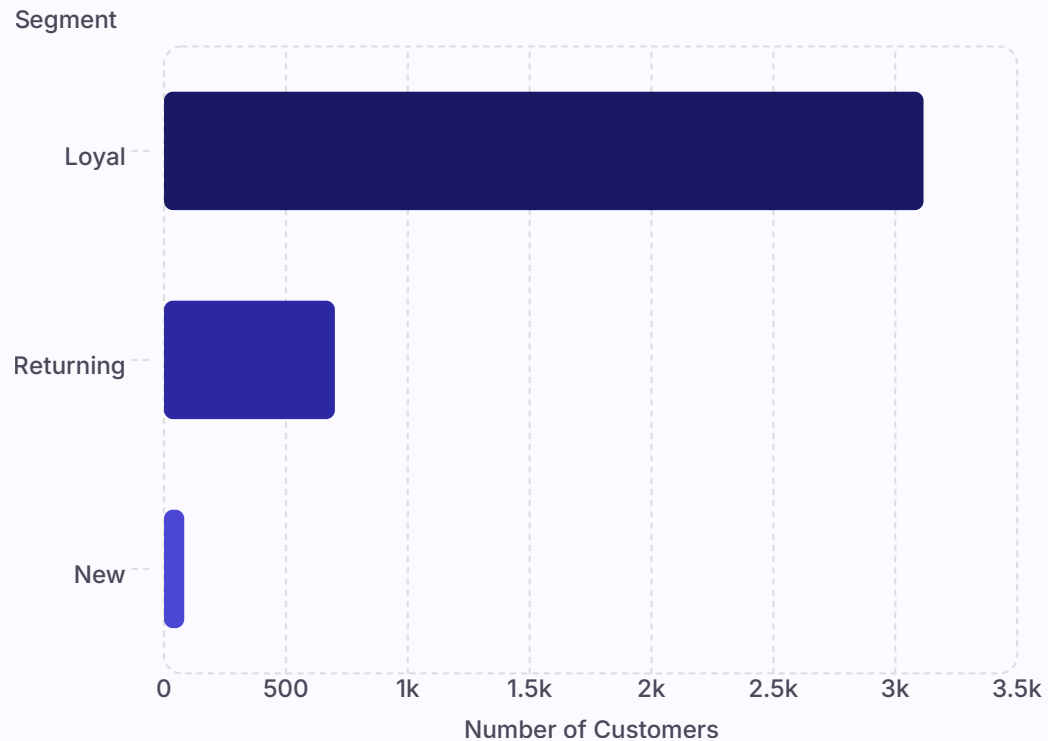
1,053

Subscribers

2,847

Non-Subscribers

Customer Segmentation



80% are Loyal — but are they growing?

Only **83 new** customers are in the dataset. Retention is strong; acquisition is the gap.

Loyal — 3,116

High-frequency, high-value

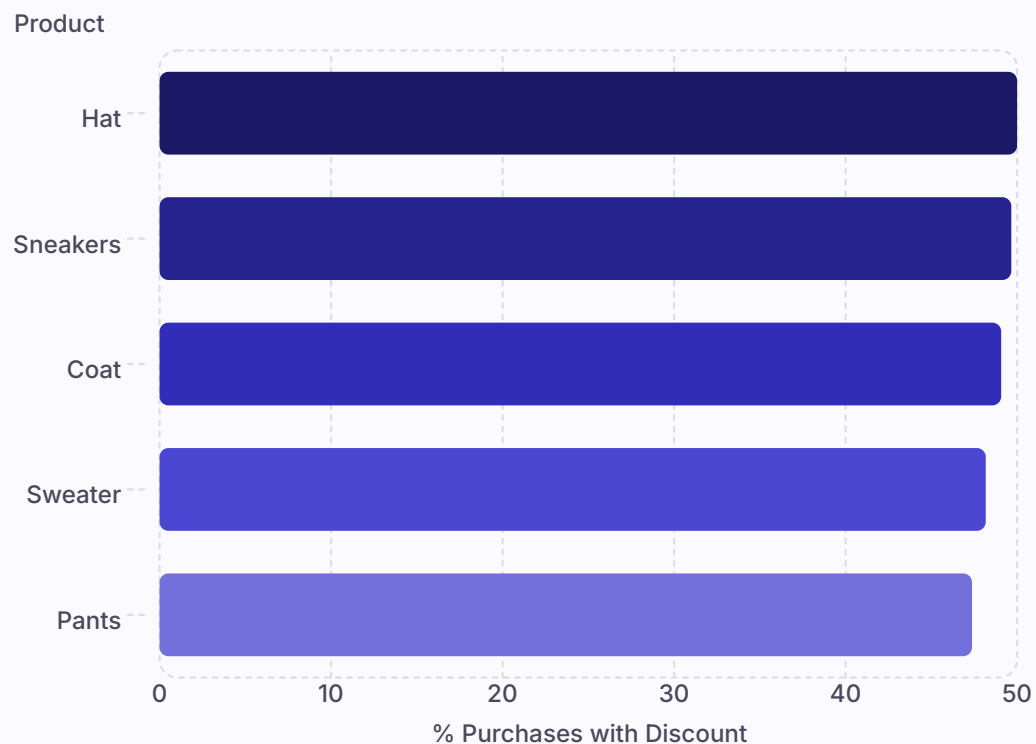
Returning — 701

Repeat buyers, not yet loyal

New — 83

First-time buyers, conversion opportunity

Discount Dependency by Product



Nearly **half of sales** rely on discounts

The top 5 products show ~47–50% discount rates. This signals price sensitivity — and margin risk.



Review discount policy: are promotions cannibalizing full-price revenue?

Top Products by Category

Accessories

1. Jewelry — 171 orders
2. Sunglasses — 161
3. Belt — 161

Clothing

1. Blouse — 171 orders
2. Pants — 171
3. Shirt — 169

Footwear

1. Sandals — 160 orders
2. Shoes — 150
3. Sneakers — 145

Outerwear

1. Jacket — 163 orders
2. Coat — 161



Power BI Dashboard



Interactive Customer Behavior Dashboard



Revenue by Category & Age Group



Filters: Gender, Subscription, Category, Shipping



27% Subscribers vs. 73% Non-Subscribers



Strategic Recommendations

Boost Subscriptions

Promote exclusive perks to convert the 73% non-subscriber base.

Loyalty Programs

Reward returning customers (701) to advance them to Loyal.

Review Discount Policy

~50% discount rate on top products risks margin erosion.

Targeted Marketing

Focus on the male segment and young adults — highest revenue contributors.